

Prerequisites

A VLAN has been created.

Configuration Impact

If the **port trunk pvid vlan** command is run more than once, the last configuration overrides the previous one.

Follow-up Procedure

Add a trunk interface to the created VLAN or VLANs.

Precautions

Default VLAN packets may not be allowed to pass through an interface. To allow them to pass through the interface, run the **port trunk allow-pass** command to add the interface to the default VLAN .

Example

```
# Set the default VLAN to VLAN 5 for interfaces.
<HUAWEI> system-view
[~HUAWEI] vlan 5
[*HUAWEI-vlan5] quit
[*HUAWEI] interface 100GE 1/0/1
[*HUAWEI-100GE1/0/1] portswitch
[*HUAWEI-100GE1/0/1] port link-type trunk
[*HUAWEI-100GE1/0/1] port trunk pvid vlan 5
```

2.4.3.36 port vlan-mapping**Function**

The **port vlan-mapping** command configures VLAN mapping on an interface.

The **undo port vlan-mapping** command deletes VLAN mapping on an interface.

By default, VLAN mapping is disabled on interfaces.

 NOTE

This command is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6820H, CE6820H-K, CE6820S, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.

Format

```
port vlan-mapping vlan vlan-id1 [to vlan-id2] map-vlan vlan-id3  
[remark-8021p 8021p-value]
```

```
port vlan-mapping vlan vlan-id1 inner-vlan vlan-id5 map-vlan vlan-id3 [map-  
inner-vlan vlan-id4] [remark-8021p 8021p-value]
```

```
port vlan-mapping vlan vlan-id1 inner-vlan vlan-id5 map-single-vlan vlan-id3
```

```
port vlan-mapping vlan vlan-id1 inner-vlan vlan-id5 to vlan-id6 map-vlan vlan-  
id3 [remark-8021p 8021p-value]
```

undo port vlan-mapping { vlan *vlan-id1* [to *vlan-id2*] [map-vlan *vlan-id3* [remark-8021p *8021p-value*]] }

undo port vlan-mapping vlan *vlan-id1* inner-vlan *vlan-id5* [map-vlan *vlan-id3* [map-inner-vlan *vlan-id4*] [remark-8021p *8021p-value*]]

undo port vlan-mapping all

undo port vlan-mapping vlan *vlan-id1* inner-vlan *vlan-id5* map-single-vlan *vlan-id3*

undo port vlan-mapping vlan *vlan-id1* inner-vlan *vlan-id5* to *vlan-id6* [map-vlan *vlan-id3*] [remark-8021p *8021p-value*]

Parameters

Parameter	Description	Value
to <i>vlan-id2</i>	Specifies a VLAN ID range in the tag carried in the received packet. <vlan-id2> specifies the end value of the VLAN ID range in the tag carried in the received packet.	The value is an integer ranging from 1 to 4094.
to <i>vlan-id6</i>	Specifies the VLAN ID before mapping. For double-tagged packets, this parameter specifies the end VLAN ID in the inner VLAN tag before mapping.	The value is an integer ranging from 1 to 4094.
map-vlan <i>vlan-id3</i>	Specifies the VLAN ID after mapping.	The value is an integer ranging from 1 to 4094.
remark-8021p <i>8021p-value</i>	Specifies the 802.1p priority in the VLAN tag to be mapped to. If this parameter is specified, after a Layer 2 Ethernet interface receives tagged packets, it maps the VLAN tag of packets to a specified VLAN tag and uses the specified 802.1p priority. If double-tagged packets are received, this parameter specifies the 802.1p priority in the outer VLAN tag to be mapped to.	The value is an integer ranging from 0 to 7.

Parameter	Description	Value
vlan <i>vlan-id1</i>	Specifies a VLAN ID range in the tag carried in the received packet. <vlan-id1> specifies the start value of the VLAN ID range in the tag carried in the received packet.	The value is an integer ranging from 1 to 4094.
inner-vlan <i>vlan-id5</i>	Specifies the VLAN ID before mapping. For double-tagged packets, this parameter specifies the start inner VLAN ID before mapping.	The value is an integer ranging from 1 to 4094.
map-inner-vlan <i>vlan-id4</i>	Specifies the VLAN ID after mapping. For double-tagged packets, this parameter specifies the inner VLAN ID after mapping.	The value is an integer ranging from 1 to 4094.
map-single-vlan <i>vlan-id3</i>	For double-tagged packets, this parameter specifies the outer VLAN ID after mapping.	The value is an integer ranging from 1 to 4094.
all	Deletes all VLAN mapping configurations.	-

Views

Layer 2 100GE interface view, Layer 2 10GE interface view, Layer 2 200GE interface view, 25GE-L2 view, 400GE-L2 view, Layer 2 50GE interface view, Layer 2 Eth-Trunk interface view, Interface group view

Default Level

2: Configuration level

Usage Guidelines

Usage Scenario

A network needs to be expanded with the growth of access users and data services, which poses the following challenges to the administrator:

- Existing and new sites assigned different VLAN IDs need to communicate.
- Inconsistent VLAN plans among sites access the public network causes VLAN ID conflicts. Therefore, VLANs at a site need to be isolated with those at another site.
- Sites that access to the public network have asymmetrical VLAN planning.

To prevent these problems, you can run the **port vlan-mapping** command to configure VLAN mapping on edge devices of the public network to map the

user VLAN tag to the public VLAN tag to implement inter-VLAN communication.

Prerequisites

The interface has been configured as a hybrid or trunk interface using the port link-type { hybrid | trunk } command.

Configuration Impact

VLAN mapping is implemented after packets are received on an inbound interface and before packets are sent out from an outbound interface. After VLAN mapping is enabled:

- Service VLAN IDs and user VLAN IDs change.
- Private network VLAN IDs and public network VLAN IDs change.

Precautions

- If VLAN mapping is used to implement communication between two VLANs, the IP addresses of devices in the two VLANs must belong to the same network segment. Otherwise, devices in the two VLANs must communicate through routes.
- If VLAN mapping is configured on an interface, modifying the mapping on the interface may cause transient interruption of traffic forwarding in the related VLAN.
- In an N:1 VLAN mapping scenario, broadcast, unknown unicast, and reserved multicast packets do not support reverse traffic restoration. Currently, only bidirectional Layer 2 unicast traffic supports this function.

Example

Configure 1 to 1 VLAN mapping to map VLAN 100 of received packets to VLAN 10 before the packets are forwarded.

```
<HUAWEI> system-view
[~HUAWEI] interface 100GE 1/0/1
[*HUAWEI-100GE1/0/1] port link-type trunk
[*HUAWEI-100GE1/0/1] port vlan-mapping vlan 100 map-vlan 10
```

Configure 2 to 2 VLAN mapping to map outer VLAN 100 of received packets to outer VLAN 10 and retain inner VLAN 200 to 300 before the packets are forwarded.

```
<HUAWEI> system-view
[~HUAWEI] interface 100GE 1/0/1
[~HUAWEI-100GE1/0/1] port link-type trunk
[~HUAWEI-100GE1/0/1] port vlan-mapping vlan 100 inner-vlan 200 to 300 map-vlan 10
```

Configure 2 to 2 VLAN mapping to map outer VLAN 100 and inner VLAN 200 of received packets to outer VLAN 10 and inner VLAN 20 before the packets are forwarded.

```
<HUAWEI> system-view
[~HUAWEI] interface 100GE 1/0/1
[~HUAWEI-100GE1/0/1] port link-type trunk
[~HUAWEI-100GE1/0/1] port vlan-mapping vlan 100 inner-vlan 200 map-vlan 10 map-inner-vlan 20
```

Configure 2 to 2 VLAN mapping to map outer VLAN 100 of received packets to outer VLAN 10 and retain inner VLAN 200 before the packets are forwarded.

```
<HUAWEI> system-view
[~HUAWEI] interface 100GE 1/0/1
```

```
[~HUAWEI-100GE1/0/1] port link-type trunk  
[~HUAWEI-100GE1/0/1] port vlan-mapping vlan 100 inner-vlan 200 map-vlan 10
```

Configure 2 to 1 VLAN mapping to map outer VLAN 200 and inner VLAN 100 of received packets to VLAN 20.

```
<HUAWEI> system-view  
[~HUAWEI] interface 100GE 1/0/1  
[~HUAWEI-100GE1/0/1] port link-type trunk  
[~HUAWEI-100GE1/0/1] port vlan-mapping vlan 200 inner-vlan 100 map-single-vlan 20
```

2.4.3.37 port vlan-stacking

Function

The **port vlan-stacking vlan** command enables VLAN stacking on an interface. That is, it configures the interface to add a VLAN tag to each received single-tagged packet.

The **port vlan-stacking untagged** command configures a device to add double VLAN tags to an untagged packet.

The **undo port vlan-stacking** command restores the default configuration.

The **undo port vlan-stacking untagged** command cancels the configuration of adding double VLAN tags to an untagged packet.

By default, VLAN stacking is not configured on interfaces.

NOTE

This command is supported only on the CE8850-SAN, CE6881H, CE6881H-K, CE8855, CE8851-32CQ4BQ, CE6866, CE6860-HAM, CE8865-SAN, CE6863H, CE6863H-K, CE6885-LL (standard forwarding mode), CE6885H, CE6885, CE6885-T, CE6863E-48S8CQ, CE6885L, CE6885-SAN, CE8851K, CE8851-32CQ8DQ-P, CE8850-HAM, CE6820H, CE6820H-K, CE6820S, CE6855-48XS8CQ, CE6866K, CE6860-SAN, CE9865, CE9855, CE8865 and CE8875.

Format

port vlan-stacking vlan *vlan-id1* [**to** *vlan-id2*] **stack-vlan** *vlan-id3*

port vlan-stacking vlan *vlan-id1* [**to** *vlan-id2*] **stack-vlan** *vlan-id3*
remark-8021p *8021p-value3*

port vlan-stacking untagged stack-vlan *vlan-id3* **stack-inner-vlan** *stackInnerVid*

undo port vlan-stacking vlan *vlan-id1* [**to** *vlan-id2*] [**stack-vlan** *vlan-id3*]

undo port vlan-stacking untagged [**stack-vlan** *vlan-id3* **stack-inner-vlan** *stackInnerVid*]

undo port vlan-stacking vlan *vlan-id1* [**to** *vlan-id2*] **stack-vlan** *vlan-id3*
remark-8021p *8021p-value3*

undo port vlan-stacking all